

Lecture 3

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Convergence Analysis and Root-Finding Methods

Outline

This lecture covers: 1. Different rates of convergence for numerical methods 2. Rigorous analysis of Newton's method 3. Weaker variants of Newton's method including: - Secant method - Steffensen's method - Method of bisection - Regula falsi

The following terms are defined in this lecture:

- Convergence Rates:
 - Linear convergence
 - Superlinear convergence
 - Sublinear convergence
 - Quadratic convergence
 - (q-linear convergence)
 - (q-superlinear convergence)

- Root-Finding Methods:
 - Newton's method (in detail)
 - Secant method
 - Muller's method
 - Steffensen's method
 - Method of bisection
 - Regula falsi

Definition (Convergence)

Let $(x_n)_{n=1}^{\infty}$ be a sequence converging to x_* . We say :

$(x_n)_{n=1}^{\infty}$ **converges** if

$$\lim_{n \rightarrow \infty} |x_n - x_*| = 0$$

Definition (Linear convergence)

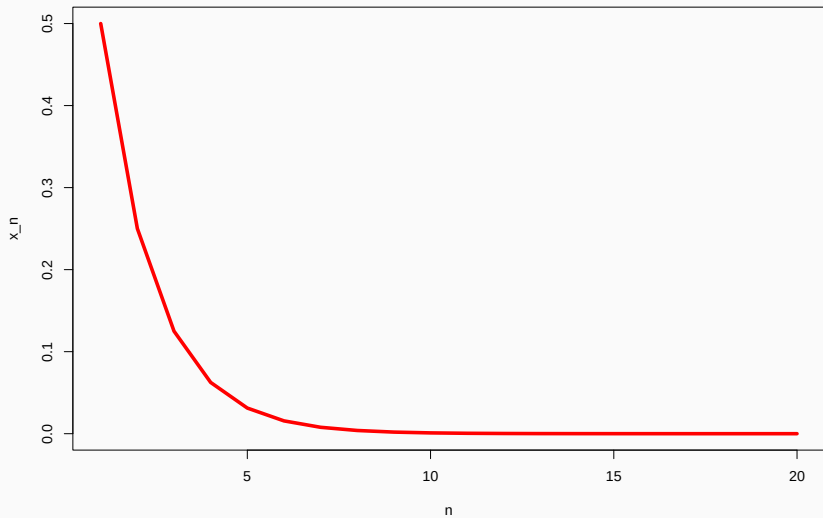
$(x_n)_{n=1}^{\infty}$ converges **linearly** if there is some $N \in \mathbb{N}$ and $0 < c < 1$ such that

$$|x_{n+1} - x_*| \leq c |x_n - x_*|$$

whenever $n > N$;

$(\gamma^n)_{n=1}^{\infty}$ exhibits linear convergence

Linear Convergence



Definition (Superlinear convergence)

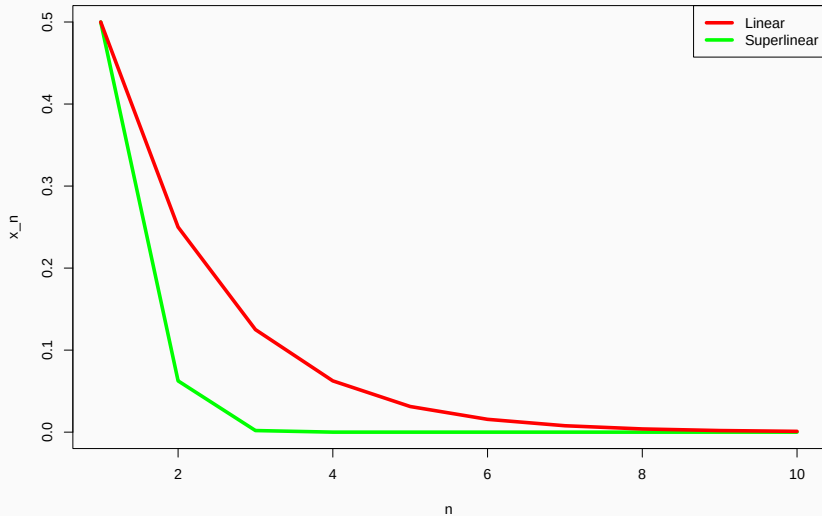
$(x_n)_{n=1}^{\infty}$ converges **superlinearly** if there is some $N \in \mathbb{N}$ and $(c_n)_{n=1}^{\infty}$ converging to 0 such that

$$|x_{n+1} - x_*| \leq c_n |x_n - x_*|$$

whenever $n > N$;

$(\gamma^{n^2})_{n=1}^{\infty}$ exhibits superlinear convergence

Superlinear Convergence



Definition (Sublinear convergence)

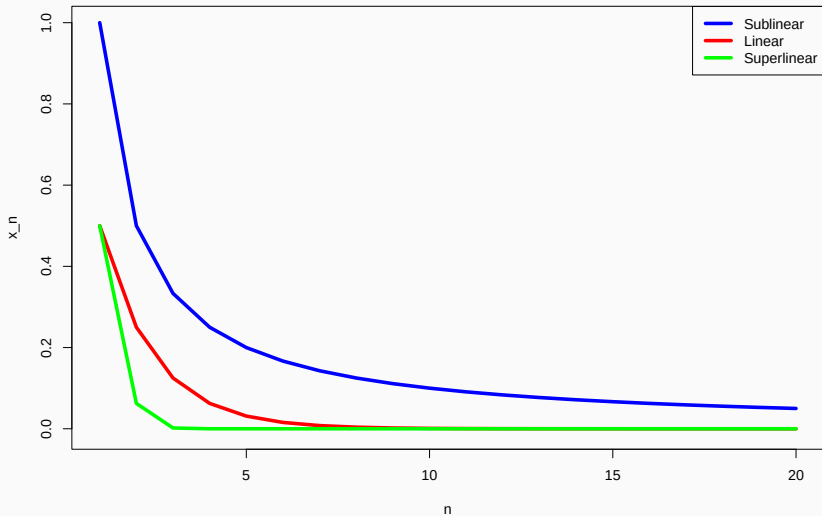
$(x_n)_{n=1}^{\infty}$ converges **sublinearly** if there is some $N \in \mathbb{N}$ and $(c_n)_{n=1}^{\infty}$ converging to 1 such that

$$|x_{n+1} - x_*| \leq c_n |x_n - x_*|$$

whenever $n > N$;

$\left(\frac{1}{n}\right)_{n=1}^{\infty}$ exhibits sublinear convergence

Sublinear Convergence



Definition (Quadratic convergence)

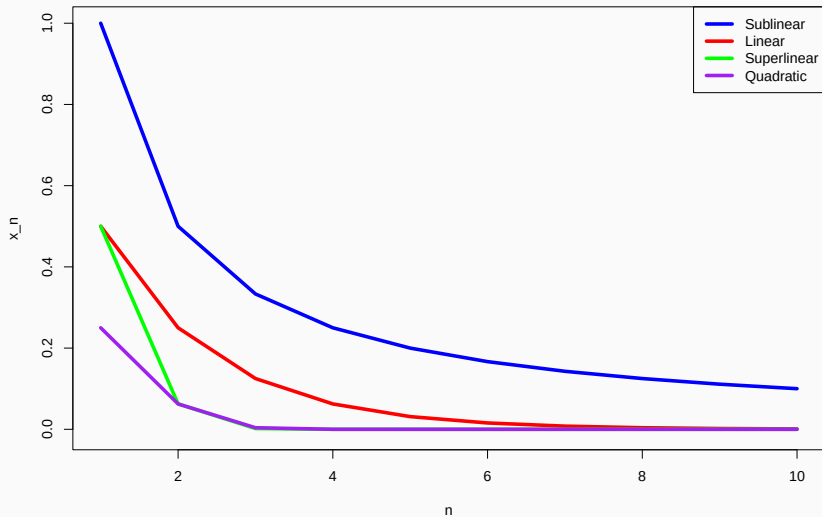
$(x_n)_{n=1}^{\infty}$ converges **quadratically** if there is some $N \in \mathbb{N}$ and $c > 0$ such that

$$|x_{n+1} - x_*| \leq c |x_n - x_*|^2$$

whenever $n > N$.

$(\gamma^{2^n})_{n=1}^\infty$ exhibits quadratic convergence

Convergence Rates



N.B. the terms " q -linearly convergent" and " q -superlinearly" are occasionally used for what we called linear and superlinear, respectively.

Higher order convergences exist (e.g., cubic convergence, with " 3 " instead of " 2 " in the quadratic convergence definition), but in practice, quadratic convergence is good enough.

Linear convergence means the error decreases by a constant factor at each step - like taking steps where each step is half the size of the previous one. This is steady but can be slow.

Superlinear convergence is faster than linear but slower than quadratic - the error reduction factor gets better with each step, but not as dramatically as in quadratic convergence.

Quadratic convergence is extremely fast - each step roughly squares the error, meaning the number of correct digits roughly doubles at each step. This is why Newton's method is so powerful when it works well.

Sublinear convergence is the slowest - the error reduction factor gets worse with each step, making it less efficient than linear convergence.

Newton's method and convergence analysis

Here is a theorem that says that if we start close enough to the solution, then Newton's method will converge quadratically to the solution:

Theorem (Newton's method)

Let $f : (a, b) \rightarrow \mathbb{R}$ and $f' \in \text{Lip}_\gamma(a, b)$. Suppose $|f'(x)| \geq \rho > 0$ for all $x \in (a, b)$. If $f(x) = 0$ has solution $x_ \in (a, b)$, then there exists some $\eta > 0$ such that if $|x_0 - x_*| < \eta$, then the sequence $(x_n)_{n=1}^\infty$ defined by*

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

converges to x_ quadratically. Specifically,*

$$|x_{n+1} - x_*| \leq \frac{\gamma}{2\rho} |x_n - x_*|^2.$$

In each step, Newton's method uses a linear approximation to $f(x)$.

We start by asking how far off this approximation can be, at the point x^* . We will use the Lipschitz condition to bound this error.

Then we ask how much this error in the function value affects the error in the root estimate. (There will be a factor of $1/f'(x_0)$ connecting the two.)

We will show that the error decreases quadratically.

We will finally need to show that after each step, we are still within our “sufficiently close” neighborhood of the solution.

Proof of Newton's Method Convergence

First, a reminder of the constants describing the function f :

- ρ is the minimum value of the derivative of f on the interval (a, b) . (Makes sure f isn't too flat.)
- γ is the Lipschitz constant of the derivative of f on the interval (a, b) . (Makes sure f' doesn't change too rapidly)
- (a, b) is the interval on which we are finding the root. We know that $f(x)$ is well behaved in this interval (slope greater than ρ , derivative Lipschitz).

We will assume there is a neighborhood around the solution x_* that is fully contained in (a, b) . Call the size of this neighborhood δ (which must be > 0):

$$(x_* - \delta, x_* + \delta) \subset (a, b)$$

We are going to show that for any *convergence rate* τ that we pick (between 0 and 1), there exists a neighborhood size η such that a) subsequent errors are at most τ times the previous error, and b) every guess is within η of the solution x_* :

$$|x_{n+1} - x_*| \leq \tau |x_n - x_*| < \eta$$

Proof. Let $0 < \tau < 1$ be fixed, and choose $\delta > 0$ be such that $(x_* - \delta, x_* + \delta) \subset (a, b)$. Put $\eta = \min\{\delta, 2\rho\tau/\gamma\}$.

In the theorem statement, we assumed that our first guess x_0 was within η of the solution x_* . We want to show what happens subsequently. So we'll start there:

$$|x_0 - x_*| < \eta.$$

Next we see what happens after one step of Newton's method. We calculate x_1 using the Newton iteration formula:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

So the error after one step is:

$$|x_1 - x_*| = \left| x_0 - x_* - \frac{f(x_0)}{f'(x_0)} \right|$$

$$= \frac{1}{|f'(x_0)|} \cdot |x_0 f'(x_0) - x_* f'(x_0) - f(x_0)|$$

Next we add in a new term $f(x_*)$ and use the fact that $f(x_*) = 0$ since x_* is the root, and rearrange the terms to get a form where we can apply Taylor's theorem.

$$= \frac{1}{|f'(x_0)|} \cdot |f(x_*) - f(x_0) - f'(x_0)(x_* - x_0)|$$

Here let's pause and show graphically what we have so far: it's an estimate of the error in the function value at x_* , divided by the slope of the linear approximation at x_0 .

Next we use Taylor's theorem to bound the error in the estimation of the function value.

$$\leq \frac{1}{|f'(x_0)|} \cdot \frac{\gamma}{2} |x_* - x_0|^2 \quad (\text{by Taylor's theorem})$$

Because the slope is greater than ρ , we can replace $1/|f'(x_0)|$ with $1/\rho$, establishing the quadratic bound.

$$\leq \frac{\gamma}{2\rho} |x_* - x_0|^2$$

We have shown that the convergence is quadratic!

We need to go a little further to show that we stay within our “sufficiently close” neighborhood of the solution at each step.

We can now use the fact that our original error was less than η :

$$\leq \frac{\gamma}{2\rho} \cdot \eta |x_* - x_0| \leq \frac{\gamma}{2\rho} |x_* - x_0|^2$$

And then we use our definition of η :

$$\leq \frac{\gamma}{2\rho} \cdot \frac{2\rho\tau}{\gamma} \cdot |x_* - x_0|$$

$$= \tau |x_* - x_0|$$

$$\leq \tau\eta < \eta$$

Since $|x_1 - x_*| < \eta$, the same argument can be repeated for $|x_2 - x_*|$, etc., to finish. Note that the τ factor is needed to guarantee strict inequality $|x_n - x_*| < \eta$ at every step.

Newton's method can fail

If we start too far from the solution, Newton's method can fail catastrophically!

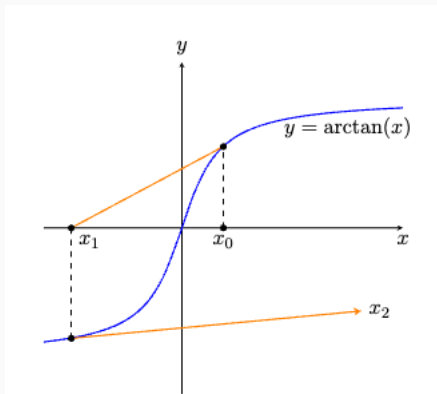


Figure 1: Newton's method applied to $\arctan(x) = 0$.

That's why need a "sufficiently close" assumption.

The condition $|x_0 - x_*| < \eta$ means that Newton's method is only quadratically convergent if we start sufficiently close to the solution x_* . In fact the method may not even be convergent if we don't have some kind of "sufficiently close" assumption.

For example, if we take $f = \arctan$, then the Newton's iteration is

$$x_{n+1} = x_n - (x_n^2 + 1) \arctan x_n.$$

If we choose $x_0 = 3$, then $x_1 = -9.4904, x_2 = 124.0007, \dots$, and $|x_n| \rightarrow \infty$.

Secant method: avoids derivatives by using only function evaluations.

Using the approximation

$$f'(x_n) \approx \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}},$$

modify Newton's method to

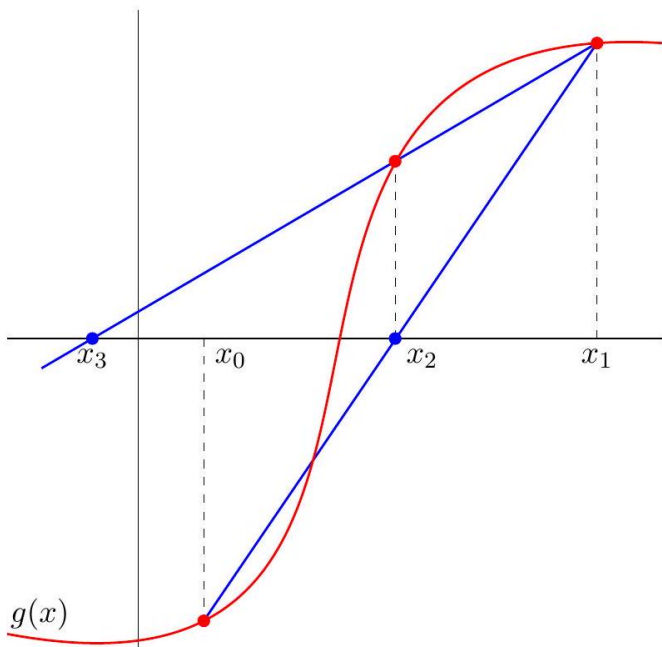
$$x_{n+1} = x_n - f(x_n) \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right].$$

It can be shown (and the proof is very similar to Theorem 2.1) that the secant method exhibits superlinear convergence:

if $f \in C^2$ and x_0, x_1 are sufficiently close to x_* , then $|\varepsilon_{n+1}| \approx c |\varepsilon_n|^\alpha$

where $\alpha = (1 + \sqrt{5})/2 \approx 1.618$ is the golden ratio, i.e., the largest real zero of $\alpha^2 - \alpha - 1 = 0$.

The “sufficiently close” assumption is crucial here as it is in Newton’s method. Figure 2 shows a divergent instance of secant method.



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\begin{algorithm}
\caption{Bisection}
\begin{algorithmic}
\Procedure{Bisection}{$a, b, \varepsilon$}
  \State  $a_0 \leftarrow a$ 
  \State  $b_0 \leftarrow b$ 
  \State  $k \leftarrow 0$ 
  \While{ $|b_k - a_k| > \varepsilon$ }
    \State  $x_k \leftarrow (a_k + b_k) / 2$ 
    \If{ $f(a_k) f(x_k) < 0$ }
      \State  $b_{k+1} \leftarrow x_k$ 
      \State  $a_{k+1} \leftarrow a_k$ 
    \Else
      \State  $a_{k+1} \leftarrow x_k$ 
      \State  $b_{k+1} \leftarrow b_k$ 
    \EndIf
  \State  $k \leftarrow k + 1$ 

```


This method relies on the Intermediate Value Theorem:

Theorem (Intermediate Value Theorem)

If $f \in C([a, b])$ and $f(a)f(b) < 0$, then there is some $x_ \in (a, b)$ such that $f(x_*) = 0$.*

Repeating this procedure, we obtain a sequence of **nested intervals**

$$[a, b] = [a_0, b_0] \supset [a_1, b_1] \supset [a_2, b_2] \supset \cdots$$

Note that

$$a_0 \leq a_1 \leq \cdots \leq b_0$$

so the sequence $(a_n)_{n=0}^{\infty}$ is monotone increasing and bounded above. Similarly, since

$$b_0 \geq b_1 \geq b_2 \geq \cdots \geq a_0$$

the sequence $(b_n)_{n=0}^{\infty}$ is monotone decreasing and bounded below. Hence $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ exist.

The length of each interval is $b_n - a_n$, and

$$b_n - a_n = \frac{1}{2} (b_{n-1} - a_{n-1}) = \cdots = \frac{1}{2^n} (b_0 - a_0),$$

so

$$\lim_{n \rightarrow \infty} (b_n - a_n) = (b_0 - a_0) \cdot \lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$$

Hence

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = x_*.$$

Let x_n be the midpoint at the n th step. Then

$$|x_{n+1} - x_*| \leq \frac{1}{2^n} |b - a|.$$

If we want an answer up to ε -precision, i.e. $|x_{n+1} - x_*| < \varepsilon$, we require

$$n = \left\lceil \log_2 \left(\frac{b-a}{\varepsilon} \right) \right\rceil.$$

Hence the method of bisection is linearly convergent.

The multivariate case is much harder since there is no n -dimensional intermediate value theorem (see Nelder-Meade, derivative-free optimization).

Regula falsi

Bisection only requires the **sign** of the function value. That's great if we can compute that efficiently!

If it happens to be just as easy to compute the function itself, we can do better using **regula falsi**.

Idea: instead of using the midpoint between a_k and b_k as our next guess, we draw a line between $(a_k, f(a_k))$ and $(b_k, f(b_k))$. Then we use the x -intercept of that line as our next guess.

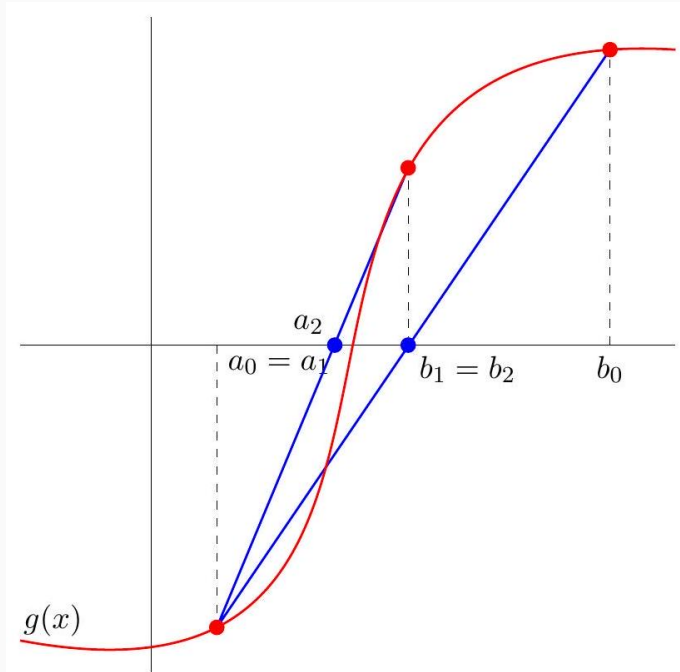


Figure 4: Pictorial view of the method of regula falsi.

The x -intercept of the line through $(a_k, f(a_k))$ and $(b_k, f(b_k))$ is given by

$$c_k = \frac{a_k f(b_k) - b_k f(a_k)}{f(b_k) - f(a_k)}$$

Regula falsi (meaning 'false position') dates back to the Babylons and Egyptians!

Why use this instead of secant method? Note that the divergent scenario in Figure 2 would not happen here - regula falsi guarantees that our iterates lie in a sequence of nested intervals (3.6) as in bisection method.

